

CSA1713 – Artificial Intelligence

Name:- B. Deepthi reddy

Reg No:- 192312288

Slot – C

5. BREADTH FIRST SEARCH (BFS)

Code:

```
print("B. Deepthi Reddy")
print(192312288)
print("Breadth First Search(BFS)")
visited = []
queue = []

graph = {}

n = int(input("Enter number of vertices: "))

for i in range(n):
    vertex = input("Enter vertex: ")
    neighbors = input("Enter adjacent vertices (space separated): ").split()
    graph[vertex] = neighbors

def bfs(start):
```

```

visited.append(start)
queue.append(start)

while queue:
    node = queue.pop(0)
print(node, end=" ")
    for i in graph[node]:
        if i not in visited:
            visited.append(i)
            queue.append(i)
start = input("\nEnter starting vertex: ")
print("BFS Traversal:")
bfs(start)

```

Output:

main.py	Output
<pre> 15 graph[vertex] = neighbors 16 17 def bfs(start): 18 visited.append(start) 19 queue.append(start) 20 21 while queue: 22 node = queue.pop(0) 23 print(node, end=" ") 24 25 for i in graph[node]: 26 if i not in visited: 27 visited.append(i) 28 queue.append(i) 29 30 start = input("\nEnter starting vertex: ") 31 32 print("BFS Traversal:") 33 bfs(start) </pre>	<pre> D. Deepika Reddy 192312288 Breadth First Search (BFS) Enter number of vertices: 6 Enter vertex: A Enter adjacent vertices (space separated): B C Enter vertex: B Enter adjacent vertices (space separated): A D E Enter vertex: C Enter adjacent vertices (space separated): A E Enter vertex: D Enter adjacent vertices (space separated): B Enter vertex: E Enter adjacent vertices (space separated): B C F Enter vertex: F Enter adjacent vertices (space separated): D E Enter starting vertex: A BFS Traversal: A B C D E F </pre>

main.py



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Output

```
1 print("B. Deepthi Reddy")
2 print(192312288)
3 print("Vacuum Cleaner Problem")
4
5 A = input("Enter Room A status (Dirty/Clean): ")
6 B = input("Enter Room B status (Dirty/Clean): ")
7
8 print("\nInitial State")
9 print("Room A =", A)
10 print("Room B =", B)
11
12 print("\nVacuum starts at Room A")
13
14 if A.lower() == "dirty":
15     print("Room A is Dirty")
16     print("Cleaning Room A...")
17     A = "Clean"
18 else:
19     print("Room A is already Clean")
```

```
B. Deepthi Reddy
192312288
Vacuum Cleaner Problem
Enter Room A status (Dirty/Clean): Clean
Enter Room B status (Dirty/Clean): Dirty

Initial State
Room A = Clean
Room B = Dirty

Vacuum starts at Room A
Room A is already Clean

Move to Room B
Room B is Dirty
Cleaning Room B...

Final State
Room A = Clean
```